

Lab Lob Book

Week - 4(Jagadeesh)

1. Model Summary

The model consists of two hidden layers with 489 and 245 neurons, derived from the last digits of my SID (2450489).

The first hidden layer has 489 neurons and the second has approximately half (245).

The model uses ReLU activation for hidden layers and one output neuron for regression.

```
model.summary()
```

Model: "sequential_1"

Layer (type)	Output Shape	Param #
dense_5 (Dense)	(None, 489)	244,989
dense_6 (Dense)	(None, 245)	120,050
dense_7 (Dense)	(None, 1)	246

Total params: 365,285 (1.39 MB)

Trainable params: 365,285 (1.39 MB)

Non-trainable params: 0 (0.00 B)

2. Training code and process

```
history = model.fit(
    X_train, y_train,
    batch_size=10,
    epochs=10,
    validation_split=0.2,
    verbose=1
)
```

```
Epoch 1/10
2640/2640 — 5s 2ms/step - loss: 0.0039 - mae: 0.0185 - val_loss: 0.0097 - val_mae: 0.0831
Epoch 2/10
2640/2640 — 4s 2ms/step - loss: 3.0982e-04 - mae: 0.0135 - val_loss: 0.0093 - val_mae: 0.0880
Epoch 3/10
2640/2640 — 4s 2ms/step - loss: 1.4753e-04 - mae: 0.0094 - val_loss: 0.0032 - val_mae: 0.0503
Epoch 4/10
2640/2640 — 4s 2ms/step - loss: 1.1532e-04 - mae: 0.0082 - val_loss: 0.0033 - val_mae: 0.0513
Epoch 5/10
2640/2640 — 4s 2ms/step - loss: 8.7876e-05 - mae: 0.0072 - val_loss: 0.0026 - val_mae: 0.0462
Epoch 6/10
2640/2640 — 4s 2ms/step - loss: 7.6408e-05 - mae: 0.0066 - val_loss: 0.0022 - val_mae: 0.0423
Epoch 7/10
2640/2640 — 4s 2ms/step - loss: 6.9146e-05 - mae: 0.0063 - val_loss: 0.0028 - val_mae: 0.0496
Epoch 8/10
2640/2640 — 4s 2ms/step - loss: 6.4206e-05 - mae: 0.0060 - val_loss: 0.0019 - val_mae: 0.0404
Epoch 9/10
2640/2640 — 4s 2ms/step - loss: 5.6918e-05 - mae: 0.0057 - val_loss: 9.4850e-04 - val_mae: 0.0268
Epoch 10/10
2640/2640 — 4s 2ms/step - loss: 5.6275e-05 - mae: 0.0057 - val_loss: 6.4016e-04 - val_mae: 0.0215
```

3. Resulting MAE

The model achieved a Mean Absolute Error (MAE) of approximately 0.0280 and a Mean Squared Error (MSE) of 0.00112 on the test dataset.

These low error values demonstrate that the MLP model performs accurately on unseen data.

```
: mse, mae = model.evaluate(X_test, y_test, verbose=0)
  print("MAE on test:", mae)
  print("MSE on test:", mse)
```

```
MAE on test: 0.02809421345591545
MSE on test: 0.0011209448566660285
```